

Electroweak Symmetry Breaking and the Matter Sector

Massive Timelike Worldlines, the Unbroken U(1), and the Composition
of β_m
in the Informational Actualization Model

Heath W. Mahaffey

Independent Researcher, Entiat, WA 98822, USA

hmahaffeyges@gmail.com

<https://github.com/hmahaffeyges/IAM-Validation>

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Abstract

In the Informational Actualization Model (IAM) the matter sector is defined by causal structure: degrees of freedom on timelike worldlines undergo gravitational decoherence and produce informational entropy, while degrees of freedom on null worldlines do not. This gives $\mu(a) < 1$ for matter perturbations and $\Sigma(a) = 1$ for light. Here we trace the particle-physics origin of that division. Electroweak symmetry breaking at $T \approx 100$ GeV ($t \approx 10^{-11}$ s) gives the $W^\pm$, Z^0 and the charged fermions their masses, placing them on timelike worldlines; the unbroken $U(1)_{\text{em}}$ keeps the photon exactly massless and on null geodesics. Within IAM the transition is therefore the epoch at which a matter sector capable of gravitational decoherence first exists, while the photon is protected from it by an exact gauge symmetry. We separate what this establishes from what it does not: photon masslessness guarantees that photons never enter the matter sector, but $\Sigma = 1$ remains a statement about how metric potentials respond to matter and is tested by weak lensing. We also write the coupling $\beta_m = \Omega_m/2$ as a sum of a baryonic part (15.6%) and a dark-matter part (84.4%), and we state the open question of whether the selection of the Higgs vacuum is itself a decoherence event.

Keywords: electroweak symmetry breaking; Higgs mechanism; virial theorem; gravitational decoherence; CP violation; baryon asymmetry; μ - Σ framework; Informational Actualization Model

1 Introduction

IAM identifies an informational contribution to horizon entropy produced by gravitational decoherence during structure formation [Mahaffey, 2026a]. The contribution enters the matter-sector perturbation equations through an enhanced Hubble friction and leaves photon propagation unchanged, so that $\mu(a) = E_{\Lambda\text{CDM}}^2(a)/E_{\text{IAM}}^2(a) < 1$ and $\Sigma(a) = 1$. The division rests on a single criterion: a degree of freedom decoheres gravitationally only if it accumulates proper time, $\Delta\tau > 0$.

That criterion has a particle-physics history. Above the electroweak scale the Standard Model gauge bosons and fermions are massless, and their excitations propagate on null geodesics. The Higgs mechanism gives them mass. This paper sets out what the electroweak transition contributes to the IAM picture, what it does not, and which statements are testable.

2 The virial theorem and the interactions

For a bound system with potential energy homogeneous of degree k in the coordinates, $V(\lambda r) = \lambda^k V(r)$, the time-averaged virial theorem gives

$$2\langle K \rangle = k \langle V \rangle. \quad (1)$$

For the $1/r$ potentials of Newtonian gravity and electrostatics, $k = -1$ and $2\langle K \rangle + \langle V \rangle = 0$ [Clausius, 1870]. This relation holds for atoms and molecules (to the precision of their computed ground states), for stars and galaxy clusters, and it is the basis of the IAM coupling $\beta_m = \Omega_m/2$ [Mahaffey, 2026b]. In IAM the kinetic half $\langle K \rangle$ is identified with the heat released irreversibly at virialisation, the Landauer cost of the quantum-to-classical transition.

The strong interaction is different. The static quark potential is well described by the Cornell form $V(r) = -\frac{4}{3}\alpha_s/r + \sigma r$, with string tension $\sigma \approx 0.18 \text{ GeV}^2$: Coulomb-like ($k = -1$) at short distance and linear ($k = +1$) at the confinement scale, where Eq. (1) gives $2\langle K \rangle = +\langle V_{\text{lin}} \rangle$. Hadrons are virialised bound states, but of their own potential, not of the $1/r$ form. The weak interaction forms no bound states at all: its range, $\sim \hbar/(M_W c) \approx 2.5 \times 10^{-18} \text{ m}$, is set by the W mass, and its role is to change flavour.

3 The weak interaction: flavour change and the discrete symmetries

The weak interaction, mediated by $W^\pm$ (80.4 GeV) and Z^0 (91.2 GeV), is the only interaction that changes quark and lepton flavour. It drives beta decay and the first step

of the proton–proton chain. It violates parity maximally [Wu et al., 1957] and violates CP through the phase of the CKM matrix [Cronin & Fitch, 1964, Cabibbo, 1963]; by the CPT theorem it therefore violates time-reversal symmetry at the level of fundamental interactions.

Two points matter here.

1. CP violation is one of the three Sakharov conditions for generating a baryon asymmetry [Sakharov, 1967]. The measured asymmetry, $\eta = (6.12 \pm 0.04) \times 10^{-10}$ [Planck Collaboration, 2020], fixes Ω_b . The CKM phase alone is too small to produce it, and in the Standard Model the electroweak transition is a smooth crossover rather than the first-order transition that electroweak baryogenesis requires; the mechanism that set η is unknown.
2. The thermodynamic arrow of time does not require CP violation. Entropy production, decoherence and Landauer’s bound hold for any interaction. What the electroweak transition adds is not irreversibility but massive particles.

4 Electroweak symmetry breaking and the IAM sectors

4.1 Before the transition

Above $T \approx 160$ GeV the Higgs expectation value vanishes in the Standard Model. The W , Z and the fermions have no rest mass (they carry thermal masses from the plasma, which do not change their light-like dispersion at high momentum). In the IAM criterion they do not accumulate proper time and do not decohere gravitationally. In any case the activation function is negligible at these epochs: $E(a) = \exp(1 - 1/a)$ is $\exp(-10^{15})$ at $T \sim 100$ GeV and only $E(z = 10) = 4.5 \times 10^{-5}$ at the onset of structure.

4.2 The transition

As the universe cools through the electroweak scale the Higgs field acquires its vacuum expectation value $v = 246$ GeV. The $W^\pm$ and Z^0 acquire mass, the charged fermions acquire masses $m_f = y_f v / \sqrt{2}$, and their excitations move on timelike worldlines with $\Delta\tau > 0$. In IAM language the matter sector exists from this epoch onward. (Dark matter, whose particle nature is unknown, joins the matter sector by the same criterion if it is massive.)

4.3 The unbroken U(1) and the photon sector

The electromagnetic U(1) survives the transition unbroken, so the photon remains exactly massless; the experimental bound is $m_\gamma < 10^{-18}$ eV [Particle Data Group, 2022]. Photons therefore stay on null geodesics, accumulate no proper time, and by the IAM criterion never produce informational entropy. This makes the photon exemption a consequence of an established symmetry rather than an extra assumption.

It does not by itself establish $\Sigma = 1$. The quantity Σ describes how the lensing potential $\Phi + \Psi$ responds to the matter distribution. In IAM $\Sigma = 1$ follows from $\delta\phi = 0$ and the absence of anisotropic stress in the informational sector [Mahaffey, 2026a]. Euclid weak lensing, with projected $\sigma(\Sigma_0) \approx 0.02$ [Euclid Collaboration et al., 2020], tests it directly.

5 The sequence of bound structures

Figure 1 and Table 1 place the transition in the standard thermal history. Times and energies are standard cosmology and the Standard Model [Particle Data Group, 2022]; the grand-unification entry is a hypothesis.

Thermal history and the IAM sectors (times and energies: standard cosmology and the Standard Model)

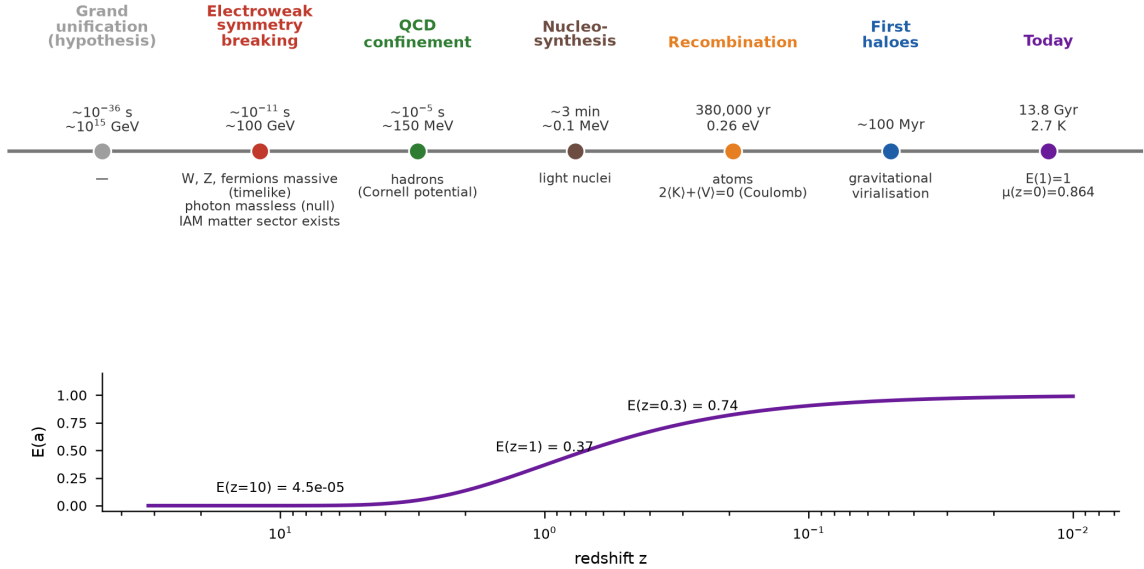


Figure 1: Thermal history and the IAM sectors. Electroweak symmetry breaking gives the massive particles of the matter sector; the photon remains massless under the unbroken U(1). Bound structures then form at increasing scales (hadrons, atoms, haloes). The activation function $E(a) = \exp(1 - 1/a)$ (purple) is negligible until the late universe, where it reaches $E(z = 1) = 0.37$ and $E(z = 0) = 1$.

Table 1: Thermal history and the formation of bound structures.

Time	Temperature	Event	Consequence
$\sim 10^{-36}$ s	$\sim 10^{15}$ GeV	Grand unification (hypothesis)	—
$\sim 10^{-11}$ s	~ 100 GeV	Electroweak symmetry breaking	W, Z , fermions massive (time-like); photon massless (null); IAM matter sector exists
$\sim 10^{-5}$ s	~ 150 MeV	QCD confinement	Hadrons: bound states of the Cornell potential
~ 3 min	~ 0.1 MeV	Nucleosynthesis	Light nuclei; Ω_b already fixed by η
380,000 yr	0.26 eV	Recombination	Atoms: $2\langle K \rangle + \langle V \rangle = 0$ (Coulomb)
~ 100 Myr	—	First haloes	Gravitational virialisation; $E(z \approx 30) \approx 10^{-13}$
~ 13.8 Gyr	2.7 K	Today	$E(1) = 1$; $\mu(z = 0) = 0.864$

6 The composition of β_m

The coupling $\beta_m = \Omega_m/2$ follows from the virial partition of gravitational energy [Mahaffey, 2026a,b]. With Planck 2018 values [Planck Collaboration, 2020], $\Omega_m = 0.3153$ and $\Omega_b = 0.0493$:

$$\beta_m = \frac{\Omega_b}{2} + \frac{\Omega_{dm}}{2} = 0.0247 + 0.1332 = 0.15765. \quad (2)$$

The baryonic part (15.6%) is fixed by the baryon asymmetry, whose origin requires CP violation and lies beyond the Standard Model (Sec. 3). The dark-matter part (84.4%) is fixed by the dark-matter density. Both are measured inputs; β_m introduces no parameter beyond them. A change in the measured Ω_m shifts β_m in proportion, $\beta_m/\Omega_m = 1/2$, which is itself testable by refitting with different Ω_m priors [Mahaffey, 2026a].

7 Open question: is vacuum selection a decoherence event?

The Higgs field settles into one point of a degenerate vacuum manifold. In a quantum treatment the early field is in a superposition over that manifold, and the emergence of a definite classical vacuum requires decoherence through coupling to the other fields of the plasma. If so, the selection of the vacuum is itself an irreversible quantum-to-classical transition, carrying a Landauer cost at the temperature of the transition. IAM does not yet treat this event, and nothing in the cosmological predictions depends on it. We state it as an open problem: whether the electroweak vacuum selection can be described within the decoherence framework of Ref. [Mahaffey, 2026a], and whether it leaves any

observable trace.

8 Tests

1. $\Sigma_0 = 0$: Euclid weak lensing at $\sigma(\Sigma_0) \approx 0.02$. A significant nonzero Σ_0 falsifies the IAM photon exemption.
2. $\mu_0 = -0.136$ ($\mu(z=0) = 0.864$): Euclid clustering, forecast detection 3.4σ (optimistic) and 5.4σ with DESI Year 5 [Mahaffey, 2026a].
3. $\beta_m/\Omega_m = 1/2$: holds for any revision of Ω_m .

9 Conclusions

Electroweak symmetry breaking gives mass to the W , Z and the charged fermions and leaves the photon massless under the unbroken $U(1)$. Within IAM this is the epoch from which a matter sector able to decohere gravitationally exists, and the photon's exact masslessness keeps it permanently outside that sector. The cosmological signature, $\mu < 1$ with $\Sigma = 1$, is set much later by structure formation and is tested by Euclid. The coupling $\beta_m = 0.15765$ is the sum of a baryonic part fixed by the unexplained baryon asymmetry and a dark-matter part, with no free parameter. Whether the selection of the electroweak vacuum is itself a decoherence event is left open.

Data Availability

Analysis scripts are available at <https://github.com/hmahaffeyges/IAM-Validation>.

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